Table 3.3 Physical properties of group 17 elements

Property	Fluorine	Chlorine	Bromine	Iodine	Astatine
Physical state at 293 K	Gas	Gas	Liquid	Solid	Solid
Atomic Number	9	17	35	53	85
Isotopes	^{19}F	^{35}Cl, ^{37}Cl	^{79}Br	^{127}I	^{210}At, ^{211}At
Atomic Mass ($g.mol^{-1}$ at 293 K)	18.99	35.45	79.9	126.9	210
Electronic configuration	$[He]2s^2\ 2p^5$	$[Ne]3s^2\ 3p^5$	$[Ar]3d^{10}\ 4s^2\ 4p^5$	$[Kr]4d^{10}\ 5s^2\ 5p^5$	$[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^5$
Atomic radius (Å)	1.47	1.75	1.85	1.98	2.02
Density ($g.cm^{-3}$ at 293 K)	1.55×10^{-3}	2.89×10^{-3}	3.10	4.93	-
Melting point (K)	53	171	266	387	573
Boiling point (K)	85	239	332	457	623

Properties:

Chlorine is highly reactive hence it doesn't occur free in nature. It is usually distributed as various metal chlorides. The most important chloride is sodium chloride which occurs in sea water.

Preparation:

Chlorine is prepared by the action of conc. sulphuric acid on chlorides in presence of manganese dioxide.

$$4NaCl + MnO_2 + 4H_2SO_4 \longrightarrow Cl_2 + MnCl_2 + 4NaHSO_4 + 2H_2O$$

It can also be prepared by oxidising hydrochloric acid using various oxidising agents such as manganese dioxide, lead dioxide, potassium permanganate or dichromate.

$$PbO_2 + 4HCl \longrightarrow PbCl_2 + 2H_2O + Cl_2$$

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

$$2KMnO_4 + 16HCl \longrightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$$

$$K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

When bleaching powder is treated with mineral acids chlorine is liberated

$$CaOCl_2 + 2HCl \longrightarrow CaCl_2 + H_2O + Cl_2$$

$$CaOCl_2 + H_2SO_4 \longrightarrow CaSO_4 + H_2O + Cl_2$$

XII U3-P-block.indd 82 04-01-2021 13:11:57